Wellbore Schematic

Operator: Aera Energy LLC
Well: 4124A-11
Type: Injector
Wellbore: 00
API no.: 0403049511

Datum: KELLY BUSHING @ 630.37ft (default)
Ground Elev: 616.37ft
PBMD: 1,306.16ft
TD MD: 1,316ft
Orientation: Directional
Completion Date: MM/DD/YYYY

CASING

Size(in)	Top MD(ft)	Base MD(ft)
14.0000	14	54
3.5000	12	1,306

CEMENT STAGES

Job Type: Primary Date: 3/13/2013

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
14.0000	24.0000	14	54

Job Type: Primary Date: 3/14/2013

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
3.5000	7.8750	12	1,303

REMARKS

Primary: 14" casing cemented at 54' to 14' with 31 sks of Unknown cement at unknown ppg, with yield of 2.71 cf/sk. Unknown cement returns to surface.

Primary: 3.5" casing cemented at 1303.16' to 12.5' with 188.3 sks of B-11 Heavy Oil Slurry TYPE III cement at 12.2 ppg, 25%wB SILICA FLOUR, 7%wB SILICALITE with yield of 2.98 cf/sk. 19.9 bbls of cement returns to surface.

MARKER DETAILS

Name	MD top(ft)
Tulare	298.73
San Joaquin/Etchegoin	1,249.37

PLUG DETAILS

Type	Top MD(ft)	Btm MD(ft)

Casing/Perfs/Slots Details	Schematic	Hole Details
Ground Level		
CSG:14.0000in, 36.71ppf, SCH 20, , 14.00ft, 54.00ft		
CSG SHOE @ 54ft		24.0000in to 54.00ft
CSG:3.5000in, 9.30ppf, K-55, EUE 8rd, 43.47ft, 794.53ft		
FLAG JOINT @ 795ft		
PERFORATED, PERFORATED, 933ft, 934ft, 1shot/ftSPF		
PERFORATED, PERFORATED, 956ft, 957ft, 1shot/ftSPF		
PERFORATED, PERFORATED, 1,000ft, 1,001ft, 1shot/ftSPF		
PERFORATED, PERFORATED, 1,076ft, 1,077ft, 1shot/ftSPF		
CSG:3.5000in, 9.30ppf, K-55, EUE 8rd, 804.63ft, 1,276.02ft		
PERFORATED, PERFORATED, 1,095ft, 1,096ft, 1shot/ftSPF		
PERFORATED, PERFORATED, 1,122ft, 1,123ft, 1shot/ftSPF		
PERFORATED, PERFORATED, 1,128ft, 1,129ft, 1shot/ftSPF		
PERFORATED, PERFORATED, 1,139ft, 1,140ft, 1shot/ftSPF		
PERFORATED, PERFORATED, 1,188ft, 1,189ft, 1shot/ftSPF		
CSG:3.5000in, 9.30ppf, K-55, EUE 8rd, 1,276.02ft, 1,306.16ft		
CSG SHOE @ 1,306ft		7.8750in to 1316.00ft

Initials: xyz Date: mm/dd/yyyy

Well Notes: